

BHOOMIKA SATHISH RAO

Electronics & Embedded Systems Engineer | Test, Verification & Validation | MSc High Commendation | Right to Work: Graduate Route Visa (No Sponsorship Required Until March 2027)

Nottingham, UK • +44 07778071365 • bhoomikasrao2k@gmail.com • [LinkedIn](#) • [GitHub](#)

PROFESSIONAL SUMMARY

Electronics and Embedded Systems Engineer with an MSc in Engineering Electronics with High Commendation from Nottingham Trent University and hands-on experience in embedded systems, electronics testing, verification and validation, sensor integration, hardware-software integration, and PCB-level testing. Practical industry exposure includes board-level functional testing, signal verification using oscilloscopes and laboratory instruments, aerospace manufacturing, and embedded automation prototyping. Experienced in Embedded C, Arduino-based microcontrollers, MATLAB, Simulink, communication protocols, structured testing, and technical documentation. Seeking early-career opportunities across electronics, embedded systems, test and validation, control systems, and related engineering roles.

CORE TECHNICAL SKILLS

Test, Verification & Validation: Structured test procedure execution, test result recording and analysis, fault diagnostics, functional and verification testing, non-conformance documentation, hardware validation, system-level validation, ISO 9001 quality environments

Embedded & Microcontrollers: Arduino UNO/Mega, Embedded C, sensor integration, firmware debugging, real-time control logic, STM32 (exposure)

Control Systems & Automation: Control systems, PID closed-loop control, sensor feedback, automation prototyping, threshold-based anomaly detection, control panel wiring

Electronics & Hardware: Analogue and digital circuit design, PCB assembly, SMD, RoHS, schematic review, hardware bring-up, signal analysis, hardware-software integration

Communication Protocols & Interfaces: I2C, UART, SPI, Bluetooth HC-05, TCP/IP, master-slave Bluetooth architecture

Engineering Tools: MATLAB, Simulink, Arduino IDE, OrCAD (basic exposure), Tinkercad

Technical Documentation: Validation reports, structured test procedures, technical documentation, engineering handover documents, MS Office Suite

Languages: English (Fluent), Hindi (Fluent), Marathi (Native), German (A2, in progress)

EDUCATION

MSc Engineering Electronics - High Commendation | *Nottingham Trent University, UK* Sept 2024

Embedded Systems • Control Systems • Digital Signal Processing • Wireless Communication

MSc project: Designed and validated a microcontroller-based system integrating sensors and actuators, with full system-level validation using laboratory instrumentation.

BEng Electronics & Communication Engineering - CGPA 7.5 | *Maharaja Institute of Technology, India* 2022

Analogue Electronics • Digital Systems • Microcontrollers • Communication Systems • Signal Processing

PROFESSIONAL EXPERIENCE

Embedded Systems Intern | *Cyient DLM Pvt Ltd, India* Jan 2023 – Apr 2023

- Supported board-level functional and verification testing within a professional electronics production environment, contributing to hardware bring-up, structured fault documentation, and accurate test result recording.
- Used oscilloscopes and laboratory instruments to perform signal verification on PCB assemblies, analyse circuit behaviour, and support structured troubleshooting within R&D and production environments.
- Worked within an ISO 9001 production environment, contributing to engineering documentation and gaining practical exposure to quality and test compliance processes.

Intern | *Aerospace Pvt Ltd, India* Apr 2022 – Jun 2022

- Supported aerospace component fabrication within a quality-controlled manufacturing environment, following structured production processes and documentation requirements.
- Contributed to process improvement activities supporting reduced inspection cycle time and assisted with cross-team documentation between engineering and production.

- Gained practical exposure to structured aerospace manufacturing workflows, quality-controlled processes, and coordination between engineering and production functions.

Embedded Prototyping Intern | *Tycoon Innovative Technology, India Sep 2021 – Nov 2021*

- Assembled and tested automation prototypes involving sensor integration, control panel wiring, and PCB assembly, supporting embedded firmware debugging and hardware-software integration within an R&D environment.
- Investigated hardware-firmware integration issues during prototype development, contributing to schematic review, firmware debugging, and structured testing from initial build through verification.
- Contributed to cross-functional prototype development discussions across hardware and software disciplines, supporting activities from initial build through testing and verification.

KEY PROJECTS

Voice-Activated Home Automation System | *MSc Final Project 2024*

- Designed and validated a microcontroller-based system from requirements capture through to verification and validation, building performance models, executing structured test cases, and producing full systems documentation.
- Developed structured test procedures and executed verification testing using laboratory instrumentation, recording and analysing results, investigating anomalous behaviour, and producing a validation report documenting system performance against requirements.

Embedded Sensor Network for Industrial Automation | *MSc Project 2024*

- Built a multi-node wireless sensor network on Arduino Mega using a master-slave Bluetooth (HC-05) architecture, integrating DHT11, MPU6050, PIR, water level, and sound sensors into a unified monitoring node with a real-time LCD display.
- Implemented threshold-based anomaly detection logic for temperature spikes, vibration, and overflow conditions, triggering buzzer and LCD alerts; verified system performance through mobile and PC-based monitoring tools and produced supporting technical documentation.

Sybil Watch IoT Security Prototype | *BEng Final Year Project 2022*

- Developed and executed a structured test and verification plan, producing test procedures, running functional and integration test cases, investigating failure modes, and generating evaluation reports.
- Achieved a 90% detection rate compared with 60% in existing RPC methods, validated under simulated attack conditions using structured functional and integration testing.
- Individually responsible for the Intrusion Detection Module, packet parsing logic, and real-time SMS alerting via GSM modem as part of a three-member team.

Traffic Light Control System with Pedestrian & Night Mode | *NTU Module: Embedded Systems 2024*

- Designed and implemented an intelligent adaptive traffic light system on Arduino UNO in Embedded C, integrating an ultrasonic sensor (SRF05) for pedestrian detection and an LDR for ambient light sensing across Regular, Pedestrian, and Night operating modes.
- Built and tested the complete circuit on a breadboard, validated functionality using Tinkercad simulation, and implemented safety features including buzzer alerts and dynamic signal timing adjustments based on real-time conditions.

CERTIFICATIONS

- PCB Design with OrCAD
- SolidWorks PCB
- Semiconductor Devices
- Project Management Foundations

RIGHT TO WORK

UK Graduate Route Visa holder with full right to work in the UK and no employer sponsorship required until March 2027. Open to relocation across the UK.